Abdul Raffay Bin Ilyas

0345-6036069 | rafayilyas65@gmail.com | Islamabad, Pakistan | LinkedIn: Abdul Raffay | GitHub: Rafay

PROFESSIONAL SUMMARY

AI/Machine Learning undergraduate with hands-on experience in full-stack development, RESTful API design, and end-to-end machine learning systems spanning computer vision, regression, and time-series forecasting. Proficient in Python, .NET, FastAPI, and TensorFlow/Keras, with project experience across desktop applications, web platforms, and deployed ML pipelines.

EDUCATION

Ghulam Ishaq Khan Institute of Engineering Sciences and Technology (GIKI) *Sep 2023 – May 2027*
Bachelor of Science in Artificial Intelligence
F.Sc. Pre-Engineering *Aug 2021 – Jun 2023*

TECHNICAL EXPERIENCE

Software Development Intern — AWT (MIS) *Jul 2025 – Aug 2025*

- Developed backend functionality for 2-3 **REST API** endpoints using the **.NET framework** in **Microsoft Visual Studio**, validated via **Swagger** to ensure accurate responses.
- Developed **frontend pages** to integrate with and support backend functionality.
- Designed UI screens and layouts in **Figma** to support **UI design** tasks.

LEADERSHIP EXPERIENCE

AI Frontier Society, GIKI *May 2025 – Present*

- Supported coordination for 4-5 **seminars** and **hackathons** hosted by the society, helping track progress and logistics.
- Contribute to a full-stack project, managing frontend and backend development, database design, and overall application integration.

PROJECTS

Food Delivery Management System *Oct 2024 – Nov 2024*

- Built a Food Delivery Management System applying core data structures, including queues for order processing, stacks for order history, and linked lists for menu and customer management.
- Implemented file handling for persistent data storage, enabling users to manage food items, place orders, and maintain customer records.

Population Prediction Model (Regression) *Nov 2024*

- Built a regression model to predict population trends from historical demographic data, achieving **over 90% prediction accuracy**.

Language Translator App – Python Desktop Application *Nov 2024 – Dec 2024*

- Built a Python desktop translation application using **Tkinter** and the **Googletrans API**, supporting 100+ languages.
- Designed a user-friendly interface with error handling for smooth and reliable translation functionality.

Handwritten Roman Numeral Classifier *May 2025*

- Implemented a **CNN (TensorFlow/Keras)** with a Tkinter GUI, achieving **95% accuracy** on handwritten Roman numeral recognition.
- Evaluated model performance using confusion matrix and precision-recall curve analysis.

GearGIK – Equipment Sharing & Rental Marketplace *Dec 2025*

- Designed and deployed a peer-to-peer equipment rental marketplace connecting owners and renters through a secure online platform.
- Developed backend services in **Python** and interactive frontend components in **JavaScript**, supporting listing management, bookings, and user accounts.

PowerGridAI – Electricity Load Shedding & Power Demand Prediction *Nov 2025 – Dec 2025*

- Built an end-to-end **machine learning** pipeline in Python to predict electricity demand, identify peak hours, and flag load-shedding risks.
- Implemented regression, classification, and time-series models, deploying the solution via **FastAPI**.
- Applied **MLOps** best practices across data processing, training, and testing workflows.

Alzheimer's Disease Detection System *May 2026*

- Built a **CNN (TensorFlow/Keras)** and transfer learning framework to classify Alzheimer's disease stages from MRI images, integrating clinical data through a fusion model, achieving **89%+ classification accuracy**.
- Developed evaluation, explainability, and deployment modules using **FastAPI** for end-to-end diagnostic support.

TECHNICAL SKILLS

Python, C++, C, SQL, Basic C#, Basic JS
.NET, Firebase, NumPy, Pandas, Matplotlib, Scikit-learn, TensorFlow, Keras, Tkinter
PostgreSQL, MongoDB
FastAPI, Swagger, GraphDB
Docker, Git/GitHub, CI/CD, VS Code, Visual Studio, Streamlit
Figma, Basic UI/UX Design, MLOps